B

BUSINESSOS

Connect. Communicate. Advertise. Manage. Grow.

PRODUCT SPECIFICATION & DEVELOPMENT BLUEPRINT

Responsive web platform | Dark navy and blue visual theme | Firebase-backed architecture

Prepared: 05 November 2026

Status: Proposed specification - not yet connected to production Firebase services

Core product idea

BusinessOS is a connected business network and operating platform. It combines business discovery, messaging, advertising, storefronts, customer relationships, transactions, team operations, and later manufacturing tools. It is not just a bookkeeping dashboard.

1. Product decisions already established

- Product name: BusinessOS (working name; check trademark and domain availability before public launch).
- Interface: responsive web platform designed for phones first but comfortable on tablets and desktop.
- Visual direction: dark navy foundation, blue actions and accents, readable light surfaces and strong contrast.
- First development target: accounts and onboarding. Business profile setup follows account creation.
- Communication: a first-party BusinessOS messaging network. Phone verification is a gate before a user starts a new one-to-one chat; verified users do not need to repeat OTP for every conversation.
- Contacts: the platform will not import or depend on a phone address book. Users and businesses are found through BusinessOS profiles and directory search. Conversations use internal user and business IDs.
- Advertising: businesses can submit media to BusinessOS campaigns. Publishing can use the BusinessOS-owned public channels and, where permitted, connected business-owned social accounts.
- Media: the campaign composer should support selecting up to 1,000 media assets in one batch, subject to per-file, storage, plan, and destination-platform rules.

Important implementation principle

The first release must be small and reliable: accounts, user profiles, business setup, and role-aware routing. Messaging, advertising, and the rest of the feature catalogue should be designed into the architecture but built in later milestones.

2. Product scope and user types

BusinessOS has two connected surfaces: (1) a public website for discovering businesses and products, and (2) a private workspace for authenticated users and business teams.

Visitor - Can browse public business profiles and public product/announcement pages, and can begin a contact flow.

Personal user/customer - Can maintain a personal profile, follow businesses, shop, and message businesses after meeting the phone-verification requirement.

Business owner/admin - Can create one or more businesses, edit business details, invite team members, and control modules.

Business employee - Can perform only the tasks granted by the business owner, such as replying to messages or managing products.

Supplier/partner - Can maintain a partner identity and communicate with businesses through accepted chat invitations.

Platform moderator/admin - Can review reported content, verification requests, campaigns, and platform abuse. Platform administration is separate from a business account.

3. Recommended technical architecture

The recommended first implementation is a responsive web application using HTML, CSS, and JavaScript modules, built with Vite. This keeps the code close to the user's current strengths while preserving a maintainable modular structure.

Frontend - Vite + vanilla JavaScript ES modules + HTML/CSS. Responsive application shell, page modules, reusable components, form validation, loading states, and accessible navigation.

Authentication - Firebase Authentication for email/password and optional Google sign-in. Firebase user UID is the stable account identity.

Database - Cloud Firestore for profiles, businesses, memberships, chat metadata, messages, campaigns, media metadata, notifications, and audit records.

Media storage - Cloud Storage for Firebase for private uploads and approved public media. Use resumable uploads, progress reporting, type/size validation, and secure storage rules.

Backend - Cloud Functions for Firebase 2nd generation (Node.js) for OTP generation/verification, privileged changes, social publishing jobs, moderation actions, webhook handling, and scheduled work.

SMS provider - Hubtel or another approved Ghana-capable SMS provider for a custom phone-verification gate. Provider credentials stay on the server, never in browser JavaScript.

Push notifications - Firebase Cloud Messaging for supported web push and later native clients; in-app notification records remain in Firestore.

Security and abuse controls - Firebase App Check where supported, Firestore/Storage Security Rules, server-side authorization, rate limits, audit logs, moderation, and secret storage.

Hosting and delivery - Firebase Hosting for the web app. GitHub for source control and a CI workflow for tests/deployments once the repo is established.

3.1 Suggested project structure

```
businessos/
  index.html                # public landing page
  package.json
  vite.config.js
  firebase.json
  firestore.rules
  storage.rules
  .env.example              # public Firebase web config placeholders only
  public/
    assets/
      brand/                # approved logo files and favicon
      images/
  src/
    main.js
    router.js
    styles/
      tokens.css            # colors, spacing, typography
      global.css
      layout.css
      components.css
    components/
      AppShell.js
      FormField.js
      Button.js
      Toast.js
      Modal.js
    pages/
      LandingPage.js
      SignUpPage.js
      LoginPage.js
      ForgotPasswordPage.js
```

```
    AccountSetupPage.js
    BusinessSetupPage.js
    DashboardPage.js
    DiscoverBusinessesPage.js  # later milestone
    MessagesPage.js            # later milestone
    CampaignComposerPage.js    # later milestone
  services/
    firebase.js
    authService.js
    profileService.js
    businessService.js
    phoneVerificationService.js # calls backend only
    searchService.js
    uploadService.js           # later milestone
    messagingService.js        # later milestone
  state/
    authState.js
    appState.js
  utils/
    validators.js
    phone.js
    formatters.js
functions/
  package.json
  src/
    index.js
    otp/phoneOtp.js
    social/publishCampaign.js
    webhooks/providerWebhooks.js
    security/rateLimits.js
tests/
  validators.test.js
  authFlows.test.js
README.md
```

3.2 Initial data model (Firestore)

Collection / path	Purpose	Access rule summary
users/{uid}	Private user account profile, display name, avatar, preferences, onboarding state.	User can read/update own safe fields; privileged fields are server-managed.
users/{uid}/private/contact	Private normalized phone number and verification metadata.	Owner can read permitted fields; client cannot mark verified or change server timestamps.
businesses/{businessId}	Business name, slug, category, description, location, public status, owner UID.	Public read only for approved profile fields; writes restricted to authorized members.
businesses/{businessId}/members/{uid}	Role and permissions inside that business.	Readable to members; role changes by owner/admin or backend.
businessSearch/{searchKey}	Searchable normalized tokens and public business identifiers.	Backend-maintained; public query exposes only public profile data.
chatRequests/{requestId}	Invitation/request to start a chat, with sender, recipient, status and timestamps.	Only participants and authorized backend can access.
conversations/{conversationId}	Conversation participant UIDs, linked business IDs, last message summary.	Only participants can read; creation validates participant membership.
conversations/{id}/messages/{messageId}	Message content, sender UID, type, createdAt, attachment metadata.	Only participants can read; sender must match authenticated UID.
campaigns/{campaignId}	Business owner, copy, media references, selected channels, approval and publishing status.	Business team can manage drafts; platform moderation controls approval.
mediaAssets/{assetId}	Storage path, owner, campaign, MIME type, byte size, status and metadata.	Owner/team access; public read only after approved publication.

Collection / path	Purpose	Access rule summary
socialConnections/{connectionId}	Provider, linked account reference, scopes, encrypted token reference and expiry.	Backend-only access to secrets; UI gets safe connection status only.
publishJobs/{jobId}	Per-platform job status, retry count, provider response ID and errors.	Business sees its own jobs; only backend writes provider results.
notifications/{notificationId}	In-app notification recipient, type, target, read state.	Recipient only, server can create.
auditLogs/{logId}	Security-sensitive actions, actor, target, timestamp and outcome.	Append-only backend writes; restricted admin reads.

4. Account system and first development target

4.1 Sign-up flow

Step 1 - Visitor opens the public landing page and selects Create account.

Step 2 - User enters display name, email address, and password, accepts the Terms and Privacy Notice, and submits.

Step 3 - Firebase Authentication creates the account and returns a stable UID.

Step 4 - The application creates users/{uid} with safe profile fields and onboardingStatus = needs_profile.

Step 5 - The user confirms email if email verification is enabled, then completes the personal profile.

Step 6 - The user chooses whether to create a business now or continue as an individual/customer.

Step 7 - If creating a business, a setup wizard collects business name, category, short description, country/region, optional city/address, and contact visibility preferences.

Step 8 - The first business is created with ownerUid and a membership record. It is not automatically marked verified.

Step 9 - The user is routed to the appropriate dashboard based on their account and business membership.

4.2 Login and recovery

- Email/password sign-in is the initial default. Google sign-in can be added as an alternative after OAuth settings are configured.
- Forgot password sends Firebase password-reset email; the app shows a neutral success message to reduce account enumeration.
- After sign-in, the app loads the user profile and memberships, then routes to onboarding or workspace.
- Admin and staff permissions are checked on the server and in Firestore Rules; hiding a button in JavaScript is not security.

5. Phone verification before starting a new chat

Recommended design

Keep Firebase Authentication for the main account (email/password or Google). Add a separate phone-ownership verification step before the first new one-to-one chat. The chat itself uses the sender and recipient BusinessOS UIDs, not phone numbers as chat IDs.

5.1 What the user experiences

Step 1 - User finds a business or person through BusinessOS search and opens the public profile.

Step 2 - User taps Message. If this account has no verified phone number, BusinessOS opens the phone-verification screen.

Step 3 - User chooses country code and enters a phone number in international E.164 format; the interface shows the intended number before sending.

Step 4 - The backend creates a one-time code and asks the configured SMS provider to deliver it.

Step 5 - User enters the code. The backend checks it and returns success or a safe failure message.

Step 6 - On success, the backend records phoneVerifiedAt and a normalized private phone number for that UID. The public profile does not expose the number unless the owner explicitly chooses to display it.

Step 7 - BusinessOS creates a chat request. The recipient receives an in-app notification and can accept, decline, or block the request.

Step 8 - After acceptance, a conversation is created using UIDs/business IDs. The number is not shown to the other participant by default.

Step 9 - The verification is reused for later chats. Ask again only if the number changes, verification expires under the policy, or a security review requires it.

5.2 Important correction: where the OTP must be generated

A browser-only JavaScript function must not generate and verify a real OTP. A user can inspect or modify browser code, and any SMS-provider secret placed in frontend JavaScript can be stolen. The browser should request an OTP; trusted backend code should generate it, store only a hash, send it through the SMS provider, and verify the submitted code.

- Backend: Node.js Cloud Function (or Cloud Run service) using a cryptographically secure random generator.
- Code length: recommend 6 digits. A 4-digit code has only 10,000 possibilities and is materially easier to guess. If a 4-digit code is retained as a product requirement, strict attempt limits and short expiry are essential.
- Expiry: 5 minutes is a reasonable starting policy; invalidate the code immediately after successful use.
- Attempts: start with at most 5 attempts per code; apply per-user, per-number, per-IP, and device abuse limits.
- Resend: use a cooldown (for example, 60 seconds) and cap daily sends; do not reveal whether a number is registered in BusinessOS.
- Storage: store a hash of the code, phone, UID, expiry, attempt count, and used status. Never store or log the plain OTP.
- SMS provider credentials: store in a server-side secret manager/environment configuration, not in public JavaScript or Firestore-readable documents.
- Provider response: a successful send request does not prove the user owns the phone. Successful entry of the delivered OTP proves control of the number at that time.

5.3 Firebase and Blaze plan facts

Firebase Phone Number sign-in SMS is pay-as-you-go (Blaze) only. Firebase currently documents a standard Authentication limit of 3,000 verification SMS per day, plus IP and per-number limits; higher limits are available through Identity Platform under its own terms and pricing. Firebase web phone authentication also uses

reCAPTCHA to reduce abuse. Phone-only authentication has security trade-offs, so it is sensible to retain email/password or another stronger sign-in option.

Cloud Storage for Firebase now requires the Blaze plan to provision or keep access to the default bucket; Google states this requirement took effect on 03 February 2026. Blaze means pay-as-you-go and does not mean every use is automatically expensive, but it does mean billing controls and budget alerts are important. See the official links in the references section.

Architecture choice for BusinessOS

Because the requested OTP is a verification gate before messaging rather than the main login method, the recommended first design is Firebase email/password authentication plus a custom backend OTP gate using a Ghana-capable SMS provider such as Hubtel. We should test delivery, pricing, sender-ID approval, and Ghana route reliability before choosing the provider. Firebase Phone Auth is a valid alternative if we later decide phone number should also be the sign-in credential.

6. Business search, chat invitations, and the contact model

- Business search is based on the BusinessOS directory, not the user's phone address book.
- Every account is identified internally by Firebase UID. Every company is identified by a separate businessId.
- Public search indexes only approved public fields: business name, slug, category, location summary, public services, and keywords.
- When a user clicks a search result, the profile page displays public business information and action buttons such as Follow, View Products, and Message.
- Message creates a chat invitation. The first message may be held as a request until the target accepts, reducing spam.
- When accepted, messages are stored under a conversation document whose participant list is UID-based. A phone number is not the recipient key.
- Phone numbers are private verification/contact data. They must not be exposed to other users by default.
- Business name search needs normalized search tokens and prefix matching; for typo tolerance, ranking, and large-scale search, use a dedicated search service later rather than loading every business document into the browser.

7. Advertising and social-media publishing model

A business owner can select up to 1,000 assets for one campaign batch, enter campaign text, select where it should appear, preview the result, and submit it. Publishing must be represented as separate jobs for each destination because each network accepts different media formats, permissions, rate limits, and review rules.

7.1 Two publishing modes

Mode A - BusinessOS-owned public channels - A business submits a campaign to the BusinessOS advertising queue. After moderation/approval, BusinessOS may publish the content from official BusinessOS-owned Facebook, Instagram, TikTok, or Snapchat channels where the channel and content rules permit it. The post must clearly attribute the advertiser and link to the business profile. This is the mode that matches the idea of using central BusinessOS accounts.

Mode B - Connected business-owned accounts - A business owner connects its own eligible social account through the provider's official authorization flow. The owner chooses the destination and grants the required permissions. BusinessOS publishes to that authorized business account, not by taking or sharing passwords.

Important reality about the central-account idea

If every submitted product is posted only from BusinessOS's own social account, the audience sees BusinessOS as the publisher, not necessarily the seller. This can build the BusinessOS brand, but it does not automatically publish to each seller's own followers. The strongest long-term product offers both central BusinessOS promotion and optional connected-account publishing.

7.2 Platform restrictions that affect planning

- TikTok Content Posting API requires an approved app scope and authorization from the target TikTok account. TikTok states unaudited clients have visibility restrictions; public posting requires passing its audit process. The API supports photo and video posting but destination rules differ.
- Instagram publishing is designed for eligible Professional (Business/Creator) accounts and requires the appropriate Meta app setup, permissions, and account authorization. Personal consumer accounts should not be assumed to be publishable through the same API path.
- Snapchat Public Profile API has content management capabilities for brand/creator profiles, but access is currently allowlist-only. It requires the appropriate Snap business account, OAuth app, and access approval.
- Facebook publishing should be implemented through the official Pages/Graph API with eligible Page access, roles, tokens, and permissions; do not assume that an app can post to arbitrary personal profiles.
- No platform promises universal one-click publishing of every media item. The composer must validate each destination and report per-channel success, failure, review, and retry status.

7.3 Campaign publishing workflow

Step 1 - Business selects up to 1,000 images/videos from device or existing media library.

Step 2 - Client creates previews and validates file types/sizes; uploads are queued with a small number of concurrent transfers.

Step 3 - Each upload reports progress and can pause, resume, or cancel when supported.

Step 4 - Backend stores file metadata and creates mediaAsset records; approved public media is served from controlled public paths/CDN delivery.

Step 5 - Business adds caption, product/service link, CTA, offer dates, intended audience, and selected channels.

Step 6 - Content moderation checks ownership, prohibited items, copyright complaints, misleading claims, and platform-specific restrictions.

Step 7 - Campaign can be sent for approval, scheduled, or published immediately to destinations for which authorization and approval already exist.

Step 8 - Each destination creates a separate publishJob record; the dashboard shows posted, pending, rejected, failed, and retry states.

Step 9 - Analytics records only available, authorized metrics; do not promise access to private audience or performance data unless the provider API permits it.

8. Media upload capacity: 1,000 assets per campaign batch

Product requirement

The campaign composer should allow a business to select up to 1,000 images/videos in one batch. This is a batch-selection target, not an unlimited-storage promise.

- Use Cloud Storage for file bytes and Firestore only for metadata; do not store image/video base64 inside Firestore documents.
- Use resumable uploads and progress events. Firebase Storage web SDK supports monitoring, pausing, resuming, and cancelling upload tasks.
- Queue uploads with controlled concurrency so a mobile phone or unstable data connection is not overwhelmed.
- Generate thumbnails and compress suitable images before upload, while retaining an original when the business needs it and has quota.
- Validate MIME type, extension, file size, video duration, dimensions, and destination-specific format before allowing publication.
- Use per-business storage quota and plan limits, plus total campaign item count. Define per-file limits only after provider and social-network rules are finalized.
- Allow a media library so a business can reuse an asset in another campaign without uploading the same file again.
- Use clear usage meters, storage warnings, and cost controls. A 1,000-file upload can create significant storage, bandwidth, and processing charges.
- One business logo is a brand asset, not the same thing as 1,000 campaign photographs. Support one primary logo plus optional logo variants in the brand kit.
- An uploaded gallery of 1,000 items does not mean 1,000 photos can be published in one post. Each social network has its own limits and supported post formats; the scheduler may need to split a campaign into multiple posts.

9. First screens and user experience specification

The initial interface should use a consistent dark-navy shell, bright-blue primary actions, clear text, and responsive layouts. Forms should be simple on a phone and use one-column fields until there is enough width for a desktop layout.

Screen 1 - Public landing page - Top bar: BusinessOS mark, Discover, Marketplace, For Businesses, Sign in, Create account. Hero: “Run your business. Reach more customers. Stay connected.” Primary CTAs: Create account and Explore businesses. Supporting sections: benefits, product modules, featured businesses, how it works, trust/security, footer. The landing page is public and does not require sign-in.

Screen 2 - Create account - Fields: full/display name, email, password, confirm password, terms/privacy consent. Optional Google sign-up. Password strength and inline validation. Show password, loading state, and safe error handling. The initial release does not ask for phone OTP during account creation unless the product decision changes.

Screen 3 - Sign in - Email and password, show password, forgot password link, optional Google sign-in, and link to create account. A signed-in user is redirected to onboarding or the correct workspace.

Screen 4 - Password recovery - Email input, send reset link, neutral confirmation message, and return to sign in. Do not confirm whether an email is registered.

Screen 5 - Personal profile setup - Display name, profile photo (optional), preferred language, country, and notification preferences. Avoid asking for unnecessary personal data.

Screen 6 - Create a business - Wizard steps: business name; category; short description; country/region/city; optional address; business contact channels; logo; public profile preview. The wizard saves a draft so the user can continue later.

Screen 7 - Workspace dashboard - Sidebar on desktop / bottom or drawer navigation on mobile. Main area shows greeting, business switcher, setup checklist, recent activity, and placeholder cards for sales, customers, messages, and stock. Empty states explain how to activate each future module.

Screen 8 - Discover businesses (planned after accounts) - Search field with results, filters for industry and location, verified status, business cards, profile preview, Follow and Message buttons. Search results contain public business information only.

Screen 9 - Business profile (planned after accounts) - Logo, name, category, verification state, description, location summary, operating hours, product catalogue, announcements, reviews, and contact actions. The private phone-verification state is not shown to other users.

Screen 10 - Phone verification gate (planned messaging milestone) - When an unverified user taps Message, show the number input and SMS consent notice, then a separate OTP input screen with masked number, resend countdown, expiry notice, change-number option, and help text.

Screen 11 - Messaging workspace (planned messaging milestone) - Desktop: conversation list, active chat, and contextual business/customer panel. Mobile: one panel at a time. Supports invitation requests, read status controls, attachments, product cards, business notes, labels, and spam/block/report actions.

Screen 12 - Campaign media studio (planned advertising milestone) - Multi-file picker, drag/drop on desktop, 1,000-item batch counter, thumbnail grid, upload queue, progress, pause/retry/cancel, caption editor, channel selection, platform-specific preview, approval status, and publish-job report.

9.1 Design tokens

Token	Proposed value	Usage
App background	#081426	Deep navy page background
Sidebar / top bar	#0B1730	Primary navigation surfaces
Action blue	#2F6BFF	Primary buttons, active states, links
Soft blue	#DCE7FF	Selected or informative surfaces
Primary text	#F5F7FB / #172033	Use light text on dark surfaces and dark text on light cards
Border	#23334D	Subtle dividers on dark surfaces
Success	#15966A	Success and completed states
Warning	#D99B2B	Warnings and attention states
Danger	#D84A4A	Destructive actions and errors

10. Development roadmap

Milestone 0 - Project foundation - Create repository, Vite setup, design tokens, responsive shell, Firebase project wiring, environment setup, security-rule baseline, and local development documentation.

Milestone 1 - Accounts (first coding target) - Implement landing, sign-up, sign-in, password reset, auth-state handling, user profile document, and first-run onboarding. Test validation, duplicate accounts, reload persistence, sign-out, and protected routes.

Milestone 2 - Business identity - Build the business setup wizard, business document, owner membership, business switcher, editable public profile, and profile visibility controls.

Milestone 3 - Search and directory - Create business search with indexes, category/location filters, profile detail page, business following, and moderation/verification workflow.

Milestone 4 - Phone verification + chat invitations - Integrate chosen SMS provider through backend, secure OTP gate, chat requests, block/report tools, and UID-based conversation creation.

Milestone 5 - Real-time messaging - Implement message send/listen, unread counters, typing/presence, attachments, push notifications, and retention/security controls.

Milestone 6 - Business media and promotion - Implement media library, 1,000-item batch workflow, campaign drafts, review process, BusinessOS-owned channel publishing, and first approved social connector.

Milestone 7 - Commerce and operations - Add catalogue, orders, stock, invoicing, payments, suppliers, staff roles, and reporting in separate modules.

Milestone 8 - Advanced/enterprise - Manufacturing, factory maintenance, quality control, AI analysis, multi-branch reporting, subscriptions, and further social integrations.

11. Acceptance criteria for the first Accounts release

- A new user can create an account with valid email and password and receives clear feedback for invalid input.
- A user can sign in, sign out, and recover a forgotten password.
- Authentication state survives a page reload and protected pages redirect correctly.
- A user profile is created exactly once per Firebase UID; the client cannot write privileged fields.
- A user can complete personal setup and choose whether to create a business.
- A user can create a business draft and the first business membership records the creator as owner.
- Business names and personal data are validated and handled with appropriate privacy defaults.
- The layout works at narrow mobile widths and desktop widths without horizontal overflow.
- No SMS secret, social access token, or admin credential appears in frontend source code.
- Firestore rules deny unauthenticated writes and unauthorized cross-user reads.

12. Security, privacy, and operational rules

- Use least privilege. An owner, admin, staff member, customer, and moderator must not share unrestricted permissions.
- Phone numbers are private. Display them only when the user explicitly chooses to publish a business contact number.
- OTP and provider tokens are never stored in frontend code, source control, public documents, or logs.
- Validate every request on the backend; never trust a role, verified flag, price, payment status, or owner UID supplied only by the browser.
- Use Firestore and Storage Rules as a security boundary, not merely UI logic.
- Keep moderation, reporting, blocking, anti-spam limits, and opt-in/opt-out controls in the design from the beginning.

- Use payment-provider webhooks and server-side verification for payments; a browser success screen is not proof of payment.
- Provide privacy notice, terms, data retention, export, account deletion, consent records, and data-protection review before public launch.
- Use budget alerts and usage monitoring for Firebase, SMS, media storage, egress, and social API operations.
- Run rules tests and emulator-based tests before deploying changes to production.

13. Decisions to confirm before implementation

- OTP length: recommended 6 digits; user initially suggested 4 digits. Confirm whether the safer 6-digit default is accepted.
- SMS provider: test Hubtel and at least one backup provider for Ghana delivery, pricing, sender-ID approval, and delivery receipts.
- Email/password first, with Google sign-in as optional later; phone number is a chat-verification attribute, not the primary user ID.
- Central advertising: define review standards, paid placement rules, attribution, and how advertiser identity appears in every BusinessOS post.
- Connected accounts: obtain platform approvals and business-owner authorization before enabling direct-to-business-account posting.
- Media policy: define per-file size, video duration, storage quota, retention, and subscription tiers; 1,000 items is the batch count, not unlimited free storage.
- Business verification: define acceptable documents, reviewer roles, data retention, and appeal process.
- Brand identity: final logo, domain, brand guidelines, and app icon remain a separate design/availability check.

14. Complete 100-feature catalogue

The following is the proposed BusinessOS feature inventory. Each item is a product capability, not a claim that it will be available in the first release. The implementation roadmap above defines the sequence.

Public website and discovery

1. Professional public homepage - Introduces BusinessOS, explains the value proposition, and directs visitors to discover, register, or sign in.

2. Business directory - Lists approved business profiles so customers and partners can find businesses without importing their phone contacts.

3. Business search - Searches business names and relevant fields such as category, services, location, and keywords; supports partial matches and filters.

4. Product discovery - Lets visitors discover products published by participating businesses.

5. Service discovery - Lets visitors find businesses that provide a particular service.

6. Industry categories - Organizes businesses by sectors such as retail, wholesale, manufacturing, agriculture, transport, and professional services.

7. Featured businesses - Highlights selected or promoted businesses on the public homepage.

8. Sponsored listings - Allows paid visibility placements with clear sponsored labels and campaign rules.

9. Customer reviews and ratings - Collects reviews from eligible customers and provides reporting and moderation for abuse or fake reviews.

10. Public announcements - Displays approved business news, operating notices, launches, and special offers.

Accounts and business identity

11. Personal user accounts - Provides a personal identity for every person using BusinessOS, independent of whether they own a company.

12. Business registration - Lets an authenticated user create a business profile with a name, category, contact details, and description.

13. Business verification - Allows a business to submit evidence for a verification review; a badge is issued only after review.

14. Multiple business ownership - Allows one user to own or administer more than one business.

15. Business logos and profile images - Stores a primary business logo and profile imagery, with a separate media library for campaign assets.

16. Business descriptions - Provides structured fields for an overview, services, products, and differentiators.

17. Business location - Stores a business address, service area, and optional map location with visibility controls.

18. Operating hours - Displays opening hours, holidays, and temporary closure notices.

19. Business contact channels - Displays business-approved phone, email, website, and social links; private personal numbers remain private by default.

20. Shareable business profile link - Gives every approved business a stable public URL and optional QR code.

Business dashboard

21. Overview dashboard - Shows key business activity in one place, based on modules the business has enabled.

22. Revenue indicators - Summarizes recorded sales and received payments over selected periods.

23. Expense indicators - Summarizes entered expenses and operating costs.

24. Profit estimate - Calculates an estimate from recorded revenue and costs, while clearly distinguishing estimates from audited accounts.

25. Customer statistics - Shows customer counts, new contacts, repeat activity, and engagement.

26. Sales performance - Shows order totals and statuses such as draft, pending, paid, fulfilled, and cancelled.

27. Inventory alerts - Highlights items at or below configured stock thresholds.

28. Recent activity feed - Shows important events performed by the owner and authorized team members.

29. In-app notifications - Collects relevant alerts for chats, order updates, payments, reviews, and administration.

30. Customizable dashboard - Lets permitted users choose useful widgets and arrange their workspace.

Messaging and communication

31. One-to-one messaging - Allows two registered BusinessOS users to communicate in a private conversation.

32. Business-to-customer chat - Lets customers contact a business and lets authorized business staff respond from a shared inbox.

33. Business-to-business chat - Lets one business send a conversation invitation to another business or its designated contact.

34. Employee messaging - Allows members of the same organization to communicate according to company permissions.

35. Website visitor chat - Provides an embeddable chat entry point so visitors can contact a business before creating a full account, subject to anti-spam controls.

36. Text and emoji messages - Supports text messages, emoji, links, and basic message formatting.

37. Image sharing - Allows users to send image files and product photos within permitted size and content rules.

38. Video and document sharing - Supports permitted videos and documents, with file validation, storage quotas, and secure access rules.

39. Voice notes - Lets users record and send short audio messages.

40. Audio and video calls - Provides a future real-time calling module, likely using a dedicated WebRTC/media service rather than Firestore alone.

Messaging productivity and business context

41. Sent and delivered indicators - Shows whether the platform accepted a message and whether it reached the recipient device/session where technically available.

42. Read receipts - Shows read status when enabled by the user and supported by the conversation settings.

43. Online and last-active status - Shows presence information with privacy controls and clear definitions.

44. Typing indicators - Shows when another participant is composing a message.

45. Message reactions - Allows participants to react to individual messages.

46. Reply to a message - Keeps replies linked to the original message for context.

47. Forward messages - Allows users to forward eligible messages while preserving attribution and abuse controls.

48. Edit and delete messages - Allows controlled editing or deletion, with timestamps and audit rules where appropriate.

49. Conversation search - Searches chat text and shared items that the current user is authorized to access.

50. Shared inbox and labels - Supports pinned, archived, muted, unread, assigned, and customer-labelled conversations.

Business community and promotion

51. Business group chats - Creates multi-member conversations for teams, partners, suppliers, or customer communities.

52. Group administrators - Lets group owners assign moderators and configure posting permissions.

53. Group invitation links - Creates revocable links or invitations for joining groups.

54. Group announcements - Allows administrators to publish important messages with optional restricted replies.

55. Group polls - Collects votes on simple group questions.

56. Business status updates - Lets businesses publish temporary updates, subject to format, expiry, and moderation rules.

57. Business channels - Provides one-to-many announcement feeds where only authorized publishers can post.

58. Business following - Lets users follow businesses and manage notification preferences.

59. Opt-in broadcast messages - Allows businesses to send announcements to consenting recipients, with unsubscribe and frequency controls.

60. Advertising centre - Creates, submits, schedules, and measures promotional campaigns with approval and policy checks.

Marketplace and commerce

61. Business storefronts - Gives each business a public catalogue or shop page.

62. Product catalogue - Stores products with titles, descriptions, SKU, price, category, availability, and media.

63. Product galleries - Supports multiple images and permitted video for each product.

64. Product variations - Represents sizes, colours, model choices, and other configurable variants.

65. Pricing and discounts - Supports price lists, sale prices, promo codes, and validity dates.

66. Shopping cart - Allows customers to collect multiple items before checkout.

67. Wishlist - Lets customers save products for later.

68. Product cards in chat - Allows a seller to share a structured product card with price and a link inside a conversation.

69. Order placement and checkout - Captures customer details, selected items, quantities, delivery choice, and order totals.

70. Payments, delivery, and returns - Connects supported payment providers and records fulfilment, tracking, cancellation, and return status.

Sales, finance, and operations

71. Point of Sale - Provides a fast checkout workflow for physical stores.

72. Quotation generator - Creates a shareable price offer for a customer.

73. Invoice generator - Creates numbered invoices with business and customer details.

74. Digital receipts - Creates receipts for recorded payments or completed sales.

75. Expense recording - Records expenses with categories, dates, notes, and receipt attachments.

76. Cash-flow overview - Summarizes recorded incoming and outgoing money.

77. Inventory management - Tracks stock quantities, movements, and stock adjustments.

78. Supplier management - Maintains supplier details, terms, and purchasing history.

79. Purchase orders - Creates and tracks orders placed with suppliers.

80. Multi-branch management - Separates branch operations while enabling consolidated reporting for authorized owners.

Employees, teamwork, and manufacturing

81. Employee profiles - Stores staff details, job role, start date, and organization membership.

82. Employee roles and permissions - Restricts what each staff member can see and do.

83. Attendance management - Records check-in and check-out events using configured rules.

84. Leave and shift management - Organizes leave requests, approvals, and employee schedules.

85. Task assignment - Assigns tasks, owners, deadlines, and completion status.

86. Project collaboration - Groups tasks, files, and discussions under a project.

87. Manufacturing management - Organizes production orders, recipes/Bills of Materials, work steps, and output.

88. Machine and equipment management - Tracks equipment registers, service intervals, and maintenance requests.

89. Quality control - Records inspections, defects, corrective action, and approval status.

90. Production cost management - Estimates or records material, labour, overhead, scrap, and finished-goods costs.

AI, analytics, security, and platform administration

91. AI business assistant - Answers questions using only business data the current user is permitted to access, with source context and safeguards.

92. AI advertisement assistant - Drafts promotional copy, captions, campaign ideas, and content variations for human approval.

93. Business performance analysis - Finds trends and unusual changes in sales, expenses, customer activity, and inventory.

94. Automated reports - Generates scheduled reports for owners and authorized managers.

95. Intelligent recommendations - Suggests follow-ups, stock replenishment, campaign improvements, and operational checks.

96. Two-factor authentication - Offers stronger account protection beyond a password.

97. Secure data management - Uses least-privilege access, security rules, server-side validation, and protected secrets.

98. Backup and recovery - Defines export, backup, recovery, and retention procedures for important business data.

99. Business verification and moderation - Reviews business identity and handles reported scams, abusive messages, and misleading content.

100. Subscription and billing management - Provides plans, usage metering, invoices, and upgrade/downgrade controls.

15. Technical references checked for this blueprint

The following official documentation pages were checked on 05 November 2026. Platform rules, quotas, prices, and API review requirements can change; re-check each provider before implementation and launch.

- Firebase Auth limits and phone SMS quotas: https://firebase.google.com/docs/auth/limits

- Firebase phone authentication for web (reCAPTCHA and security considerations): https://firebase.google.com/docs/auth/web/phone-auth
- Firebase pricing plans: https://firebase.google.com/pricing
- Cloud Storage for Firebase web uploads (resumable uploads and progress): https://firebase.google.com/docs/storage/web/upload-files
- Cloud Storage for Firebase Blaze requirement (effective 03 February 2026): https://firebase.google.com/docs/storage/faqs-storage-changes-announced-sept-2024
- Hubtel developer portal (payments and developer APIs): https://developers.hubtel.com/
- Hubtel SMS API documentation: https://docs-developers.hubtel.com/
- TikTok Content Posting API - Direct Post: https://developers.tiktok.com/docs/en/content-posting-api-get-started
- TikTok Content Sharing Guidelines: https://developers.tiktok.com/docs/en/content-sharing-guidelines
- Meta Instagram API collection and professional-account publishing overview: https://www.postman.com/meta/instagram/documentation/6yqw8pt/instagram-api
- Meta Instagram content publishing documentation: https://developers.facebook.com/docs/instagram-platform/content-publishing/
- Meta Pages API documentation: https://developers.facebook.com/docs/pages-api/
- Snapchat Public Profile API introduction: https://developers.snap.com/marketing-api/Public-Profile-API/Introduction
- Snapchat Public Profile API access and allowlisting: https://developers.snap.com/marketing-api/Public-Profile-API/GetStarted

16. Immediate next build step

Start coding with Milestone 1

Create the Vite project and implement the public landing page, Sign Up, Sign In, Forgot Password, authentication state, and profile onboarding. Do not build the full messaging system yet. Keep the future phone-verification gate as a planned backend service and use UID-based identities in the data model from day one.